

**NANYANG
TECHNOLOGICAL
UNIVERSITY**

SINGAPORE

SC/CE/CZ2002: Object-Oriented Design & Programming

ASSIGNMENT

Building an OO Application

Lab Group	SCEA
Team	1
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Design Considerations

Layered Architecture

Our Internship Placement Management System is structured around three main layers. The app layer consists of the Main and different Menu classes to handle user interaction. The 2nd layer consists of the core business objects such as User, Student, CompanyRepresentative, CareerCentreStaff, Internship and Application to encapsulate the business logic and data. The third layer consists of the helper functions such as FileIOHandler, and ApplicationViewer for displaying domain object data and adhering to the Separation of concerns. This separation into the different functions was driven by object oriented concepts such as abstraction and encapsulation, and even Single Responsibility Principle (SRP). This was so that the system would be easier to extend and maintain, without rewriting and affecting the main source code.

Application of OOP principles

Firstly, **encapsulation** is rigorously applied across all entity classes, such as User, Internship, and Application, where internal state is kept private and exposed only through controlled public getter and setter methods, preserving data integrity by allowing internal implementation to change without affecting other parts of the system

Abstraction is also used through the “User” class. In the “User” class, the specific implementation details of how different actors in the system perform actions like logging in are hidden, instead only defining the essential attributes and behaviours common to all users.

The **SRP** is also applied through the way responsibilities are divided across the different classes. Each class has a focused role. For example, “MainMenu” handles login and registration. “Studentmenu” handles only student-specific interfaces. “StaffMenu” focuses on staff-specific interfaces. By clearly distinguishing and applying only one class per role, we’re able to apply SRP.

Use of Relationships and Patterns

Alongside inheritance, the system makes extensive use of composition and aggregation to model relationships between domain objects. An “Internship” aggregates a “CompanyRepresentative” and “Application”. “Application” in turn holds references to both the “Student” and the “Internship”, i.e. association. These are classic “has-a” relationships: for example, a company representative “has” internships they created, and an internship “has” many applications.

Extensibility and Maintainability

The design decision above contributes directly to the extensibility and maintainability of the system. For example, to introduce new user types, one simply needs to create a new class extending User and a corresponding menu class for it, leaving the core entity and utility untouched. Changes are isolated, like modifying the file storage format requires changes within the FileIOHandler only.

Trade-Offs

However, these choices come with some tradeoffs as well. For simplicity, the system uses static in “Main”. While this simplifies data access and is pragmatic for an application of this scale, it

introduces tight coupling and violates the Dependency Inversion Principle (DIP), making the system less amendable to testing and potentially harder to modify if the data source changes. Similarly, the direct instance of checks, while clear and straightforward, are less extensible than a fully polymorphic solution for adding new user types. Ultimately, these decisions were conscious trade-offs that prioritised implementation simplicity and code clarity at this stage of development, resulting in a design that is logically structured and maintainable, with clear pathways for future refactoring should the system's complexity grow.

Additional Features/ Functionalities

1. Forgot & Reset Password

What it does

Allows all users to reset their password if they forget their current one. Then allows creation of a new password that meets specified constraints.

Why it is beneficial

- Prevents users from being locked out of their accounts
- Improves usability and user satisfaction
- Mimics real-world account recovery systems
- Enhances security by enforcing password requirements

2. Internship Application History

What it does

Students can view all of their past internship applications, including:

- Submitted applications
- Accepted / rejected results
- Withdrawn applications
- Applications from previous sessions (persistent across runs)

Why it is beneficial

- Gives students full visibility over their academic/internship journey
- Helps them track progress and decisions
- Provides realistic functionality found in actual university portals
- Ensures data persistence across program sessions

3. Bulk Internship Creation

What it does

Allows company representatives to upload a CSV or text file containing many internship listings.
The system automatically:

- Parses the file
- Creates internships in batch
- Validates each entry before saving

Why it is beneficial

- Saves time for large companies that offer multiple similar roles
- Reduces repetitive manual data entry
- Supports scalability and real-world company needs
- Increases efficiency for power users

4. Applicant Ranking Tool

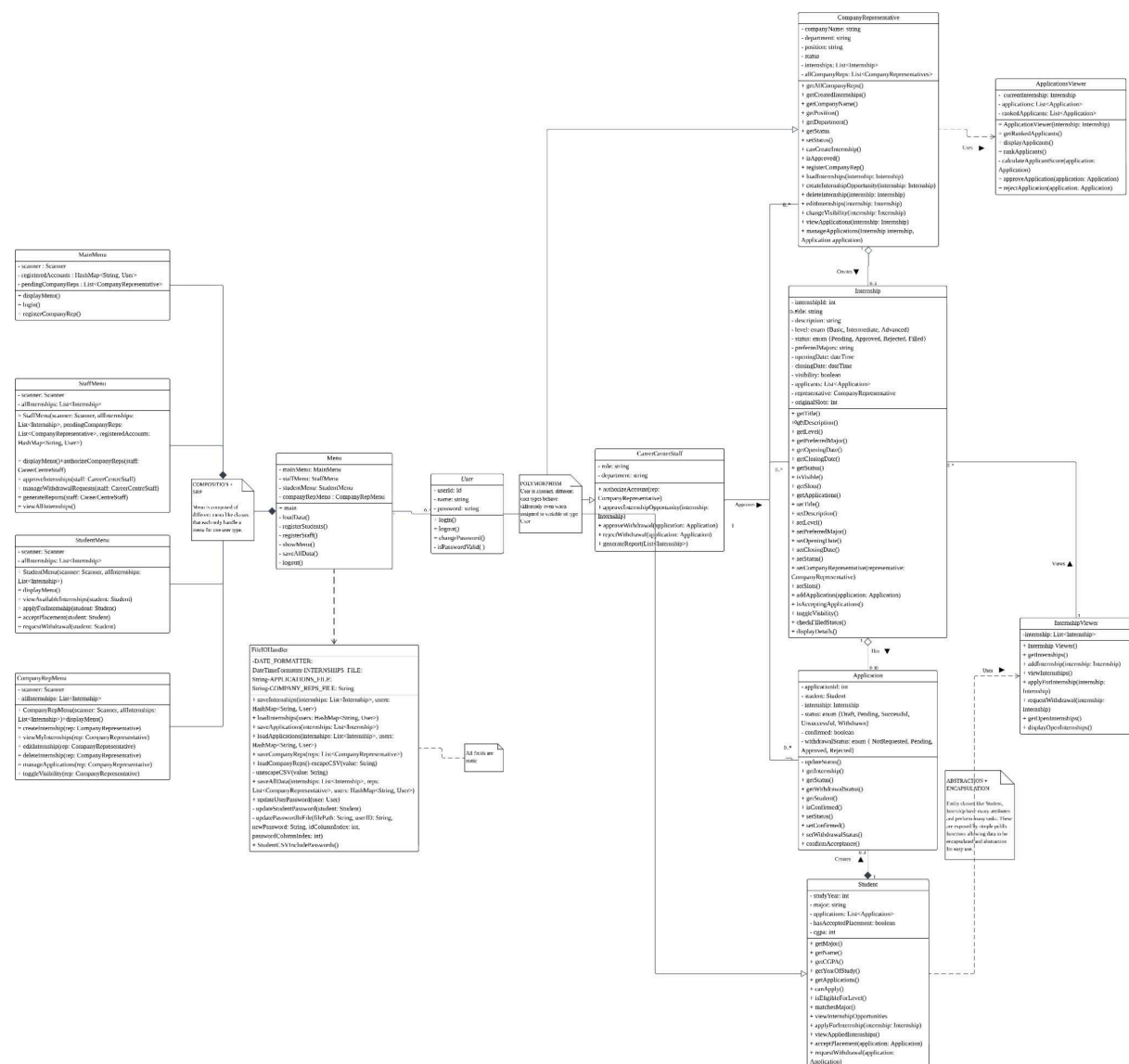
What it does

Shows sorted list of student applications by criterias of GPA and year of study for Company Reps

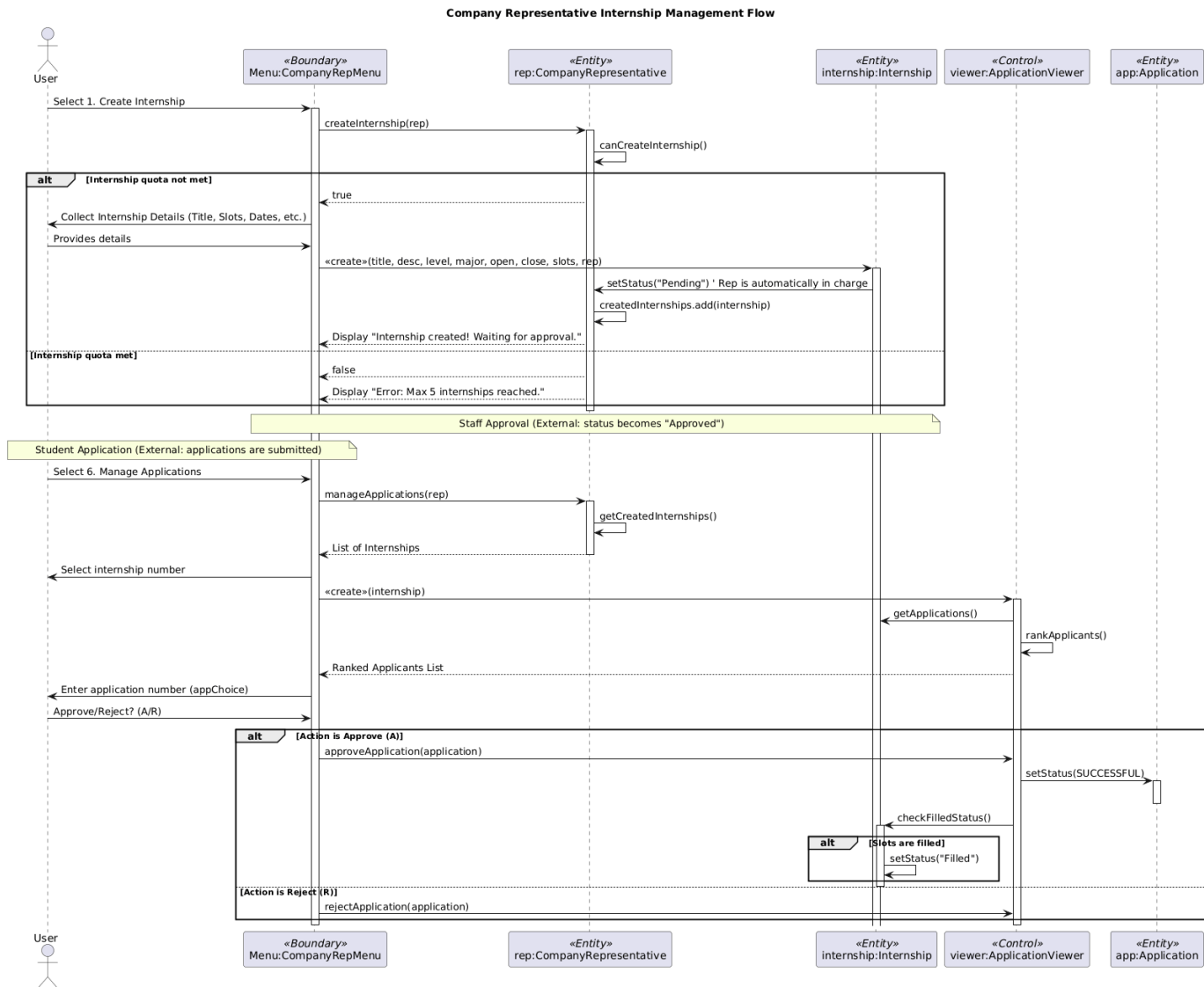
Why it is beneficial

- Helps companies prioritize applicants efficiently
- Supports various evaluation styles (academic strength and experience)
- Enhances decision-making transparency
- Provides a realistic HR-like tool inside the system

UML Class Diagram



UML Sequence Diagram



Testing

No.	Test Case	Expected Behavior	Failure Indicators
1	Valid User Login	User should be able to access their dashboard based on their roles	User cannot log in or receive incorrect error messages
2	Invalid ID	User receives a notification about incorrect ID	User is allowed to log in with an invalid ID or fail to provide

			meaningful error message
3	Forgot / Reset Password	User should be able to reset their password if they forgot it	User logs in successfully with the old password after resetting
4	Incorrect Password	System should deny access and alert the user to incorrect password	User logs in successfully with a wrong password or fail to provide meaningful error message
5	Password Change Functionality	System updates password, prompt re-login and allows login with new credentials	System does not update the password or denies access with the new password
6	Company Representative Account Creation	A new Company Representative should only be able to log in to their account after it has been approved by a Career Center Staff	Company Representative staff can log in without any authorization
7	6. Internship Opportunity Visibility Based on User Profile and Toggle	Internship opportunities are visible to students based on their year of study, major, internship level eligibility, and the visibility setting	Students see internship opportunities not relevant to their profile (wrong major, wrong level for their year) or when visibility is off
8	7. Internship Application Eligibility	Students can only apply for internship opportunities relevant to their profile (correct major preference, appropriate level for their year of study) and when visibility is on	Students can apply for internship opportunities not relevant to their profile (wrong major preference, Basic-level students applying for Intermediate/Advanced opportunities) or when visibility is off
9	Viewing Application Status after Visibility	Students continue to have access to their application	Application details become

	Toggle Off	details regardless of internship opportunities' visibility	inaccessible once visibility is off
10	Single Internship Placement Acceptance per Student	System allows accepting one internship placement and automatically withdraws all other applications once a placement is accepted	Student can accept more than one internship placement, or other applications remain active after accepting a placement
11	Company Representative Internship Opportunity Creation	System allows Company Representatives to create internship opportunities only when they meet system requirements	System allows creation of opportunities with invalid data or exceeds maximum allowed opportunities per representative
12	Internship Opportunity Approval Status	Company Representatives can view pending, approved, or rejected status updates for their submitted opportunities	Status updates are not visible, incorrect, or not properly saved in the system
13	Internship Detail Access for Company Representative	Company Representatives can always access full details of internship opportunities they created, regardless of visibility setting	Opportunity details become inaccessible when visibility is toggled off for their own opportunities
14	Restriction on Editing Approved Opportunities	Edit functionality is restricted for Company Representatives once internship opportunities are approved by Career Center Staff	Company Representatives are able to make changes to opportunity details after approval
15	Student Application Management and Placement Confirmation	Company Representatives retrieve correct student applications, update slot availability accurately, and correctly confirm placement details	Incorrect application retrieval, slot counts not updating properly, or failure to reflect placement confirmation details accurately

16	Internship Placement Confirmation Status Update	Placement confirmation status is updated to reflect the actual confirmation condition	System fails to update or incorrectly records the placement confirmation status
17	Create, Edit, and Delete Internship Opportunity Listings	Company Representatives should be able to add new opportunities, modify existing opportunity details (before approval by Career Center Staff), and remove opportunities from the system	Inability to create, edit, or delete opportunities or errors during these operations
18	Career Center Staff Internship Opportunity Approval	Career Center Staff can review and approve/reject internship opportunities submitted by Company Representatives	Career Center Staff cannot access submitted opportunities for review, approval/rejection actions fail to update opportunity status, or approved opportunities do not become visible to students as expected
19	Toggle Internship Opportunity Visibility	Changes in visibility should be reflected accurately in the internship opportunity list visible to students	Visibility settings do not update or do not affect the opportunity listing as expected
20	Career Center Staff Internship Opportunity Management	Withdrawal approvals and rejections are processed correctly, with system updates to reflect the decision and slot availability changes	Incorrect or failed processing of withdrawal requests or slot counts not updating properly
21	Generate and Filter Internship Opportunities	Accurate report generation with options to filter by placement status, major, company, level, and other specified categories	Reports are inaccurate, incomplete, or filtering does not work as expected
22	Application Withdrawal by Student	Students can withdraw their applications before approval and slots become available	Withdrawal fails or slot count doesn't update after withdrawal

		again	
23	Maximum Applications per Student	System enforces maximum number of active applications per student	Students can submit unlimited applications beyond the limit
24	Application Deadline Enforcement	System prevents applications after internship closing date	Students can apply to expired internships
25	View Application History	Student can see all past applications from current and previous sessions	History is empty after restart or shows only current session data
26	Application Status Tracking	History shows correct status (Pending, Approved, Rejected) for each application	Status displays incorrectly or doesn't update with changes
27	CSV File Import	System successfully imports 5 internships from valid CSV format	System rejects valid CSV or imports incorrect number of internships
28	Field Mapping	CSV columns correctly map to internship object fields	Data appears in wrong fields or some fields remain empty

Test Cases 1-20 were provided

Reflection

The difficulties encountered and the way to conquer, the knowledge learnt from this course, further improvement suggestions. Strong demonstration of learning points and insights of good design and implementation practices, based on experience gained from doing the assignment.

Working on this project really drove home the importance of SOLID principles and good OOP design practices—particularly Single Responsibility and Open/Closed principles. We learned the hard way that without consciously applying these guidelines, it's surprisingly easy to slip into poor coding habits.

Early on, we created several "god classes" packed with too many methods. At the time, it seemed more convenient to just tack on another method to an existing class rather than create a new one. This approach came back to bite us when the codebase became confusing and harder to navigate. Similarly, with the Open/Closed Principle, we kept falling into the trap of modifying existing, functional code to accommodate new features instead of extending functionality through new implementations. While this felt like the quick solution, it frequently introduced bugs and unexpected behavior in other parts of the application that relied on those modified methods.

File organization was another stumbling block. Initially, we dumped everything at the root level because it felt simpler. But as the project grew, this became messy fast. Once we reorganized code into proper packages and cleaned up our imports, the codebase became significantly more maintainable and we ran into fewer cross-platform compilation issues.

Beyond the technical side, team coordination proved just as crucial. Regular meetings and thoughtful task division helped us avoid messy merge conflicts. Keeping everyone updated on changes—especially when someone pushed significant updates—was essential to prevent teammates from working off outdated branches. We learned that good communication isn't just nice to have; it's necessary to keep a collaborative project running smoothly.

Looking ahead, we'd benefit from establishing better design patterns upfront and perhaps doing more frequent code reviews to catch issues earlier. The experience has definitely changed how we'll approach future projects—sometimes the "easier" path in the moment creates more work down the line.

Github Repo

https://github.com/Riccardo-gt/SC2002_Group1.git

Declaration of Original Work for SC2002 Assignment

We hereby declare that the attached group assignment has been researched, undertaken, completed, and submitted as a collective effort by the group members listed below.

We have honored the principles of academic integrity and have upheld Student Code of Academic Conduct in the completion of this work.

We understand that if plagiarism is found in the assignment, then lower marks or no marks will be awarded for the assessed work. In addition, disciplinary actions may be taken.

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Important notes:

1. Name must EXACTLY MATCH the one printed on your Matriculation Card.
2. Student Code of Academic Conduct includes the latest guidelines on usage of Generative AI and any other guidelines as released by NTU.